

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2022

TEST DATE: 23-12-2021

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer. There is no negative marking.

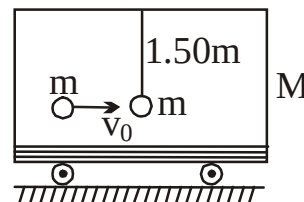
Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A ball of mass $m = 1\text{kg}$ is hung vertically by a thread of length $\ell = 1.50\text{ m}$. Upper end of the thread is attached to the ceiling of a trolley of mass $M = 4\text{kg}$. Initially, trolley is stationary and it is free to move along horizontal rails without friction. A shell of mass $m = 1\text{kg}$, moving horizontally with velocity $v_0 = 6\text{ ms}^{-1}$, collides with the ball and gets stuck with it. As a result, thread starts to deflect towards right. Calculate its maximum deflection with the vertical. ($g = 10\text{ ms}^{-2}$)



- (A) 37°
(B) 53°
(C) 30°
(D) 60°

Ans. A

Sol. When shell strikes the ball and gets stuck with it, combined body of mass $2m$ starts to move to the right. Let velocity of combined body (just after collision) be v_1 .

According to law of conservation of momentum,

$$(m + m) v_1 = mv_0 \text{ or } v_1 = \frac{v_0}{2} = 3\text{ ms}^{-1}.$$

As soon as the combined body starts to move rightwards, thread becomes inclined to the vertical. Horizontal component of its tension retards the combined body while trolley accelerates rightwards due to the same component of tension.

Inclination of thread with the vertical continues to increase till velocities of both (combined body and trolley) become identical or combined body comes to rest relative to trolley. Let velocity at that instant of maximum inclination of thread be v . According to law of conservation of momentum $(2m + M) v = 2m \cdot v_1$ or $v = 1\text{ ms}^{-1}$

During collision of ball and shell, a part of energy is lost. But after that, there is no loss of energy. Hence, after collision, kinetic energy lost is used up in increasing gravitational potential energy of the combined body.

If maximum inclination of thread with the vertical be θ then according to law of conservation of energy,

$$\begin{aligned} \frac{1}{2} (2m) v_1^2 - \frac{1}{2} (2m + M) v^2 \\ = 2mg (\ell - \ell \cos \theta) \end{aligned}$$

$$\therefore \cos \theta = 0.8 \text{ or } \theta = 37^\circ$$

2. The torque ($\vec{\tau} = \vec{r} \times \vec{F}$) of a force $\mathbf{F} = -3\hat{i} + \hat{j} + 5\hat{k}$ acting at the point $\mathbf{r} = 7\hat{i} + 3\hat{j} + \hat{k}$ is -

- (A) $14\hat{i} - 38\hat{j} + 16\hat{k}$
 (B) $4\hat{i} + 4\hat{j} + 6\hat{k}$
 (C) $-21\hat{i} + 4\hat{j} + 4\hat{k}$
 (D) $-14\hat{i} + 34\hat{j} - 16\hat{k}$

Ans. A

Sol. The torque is defined as

$$\begin{aligned}\vec{\tau} &= \mathbf{r} \times \mathbf{F} \\ \mathbf{r} \times \mathbf{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 3 & 1 \\ -3 & 1 & 5 \end{vmatrix} \\ &= \hat{i} \begin{vmatrix} 3 & 1 \\ 1 & 5 \end{vmatrix} + \hat{j} \begin{vmatrix} 1 & 7 \\ 5 & -3 \end{vmatrix} + \hat{k} \begin{vmatrix} 7 & 3 \\ -3 & 1 \end{vmatrix} \\ &= \hat{i}(15 - 1) + \hat{j}(-3 - 35) + (7 - (-9))\hat{k} \\ &= 14\hat{i} - 38\hat{j} + 16\hat{k}\end{aligned}$$

3. In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-5} cm. If the distance between the slits is 10^{-3} m, calculate the wavelength of the light used.

- (A) 4500 Å
 (B) 3000 Å
 (C) 9000 Å
 (D) 6000 Å

Ans. D

Sol. The fringe-width β is given by

$$\beta = \frac{\lambda D}{2d} \quad \text{and} \quad \beta' = \frac{\lambda D'}{2d}$$

Where λ is the wavelength of light used, D is the distance of the screen from the two slits and $2d$ is the separation between two slits.

$$\text{Now} \quad \beta - \beta' = \frac{\lambda(D - D')}{2d}$$

$$\Rightarrow \lambda = \frac{(\beta - \beta')2d}{D - D'}$$

$$\lambda = \frac{(3 \times 10^{-5})(10^{-3})}{5 \times 10^{-2}}$$

$$= 0.6 \times 10^{-6} \text{ m} = 6000 \text{ \AA}$$

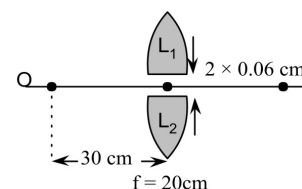
4. A certain physical quantity is calculated from the formula $\frac{\pi}{3} (a^2 - b^2) h$, where h , a and b are all length. The quantity being calculated is-

- (A) Velocity
(B) Length
(C) Area
(D) Volume

Ans. D

Sol. Given quantity is $= (a^2 - b^2) h$
 dimension of $h = [L]$
 dimensions of $a^2 - b^2 = [L^2 - L^2] = L^2$
 Therefore the dimensions of the given quantity are $[L^3]$. Thus the quantity being measured is volume

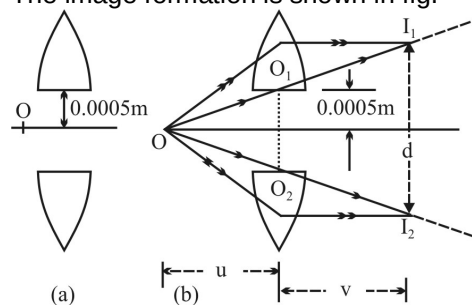
5. A point object O is placed at a distance of 0.3 m from a convex lens (focal length 0.2 m) cut into two halves each to which is displacement by 0.005 m as shown in fig. Find the position of the image. Find the distance between the images.



- (A) 0.003 m
(B) 0.002 m
(C) 0.004 m
(D) 0.001 m

Ans. A

Sol. The image formation is shown in fig.



The rays through optical centres O_1 and O_2 pass undeflected. The image of O are formed at I_1 and I_2 due to upper and lower lenses respectively.

The number of images formed = two

For a convex lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Here $u = -0.3$ m and $f = 0.2$ m

$$\therefore \frac{1}{v} = \frac{1}{u} + \frac{1}{f} = -\frac{1}{0.3} + \frac{1}{0.2} = \frac{1}{0.6}$$

$u = 0.6$ m

Let d be the distance between the two images I_1 and I_2 . From $\triangle OO_1O_2$ and $\triangle II_1I_2$, we have

$$\frac{d}{O_1O_2} = \frac{u+v}{u}$$

$$\text{or } \frac{d}{0.0005 + 0.0005} = \frac{0.3 + 0.6}{0.3}$$

Solving we get $d = 0.003$ m

6. A truck starts from rest with an acceleration of 1.5 metre/sec^2 while a car 150 metre behind starts from rest with an acceleration of 2 metre/sec^2 . How long will it take before both the truck and car to be side by side, and how much distance is travelled by each?

- (A) 2.45 sec
(B) 5 sec
(C) 24.5 sec
(D) 5.3 sec

Ans. C

Sol. Let x be the distance travelled by the truck when both truck and car are side by side. The distance travelled by the car will be

$(x + 150)$ as the car is 150 metre behind the truck. Applying the formula

$s = ut + (1/2)at^2$, we have

$$x = \frac{1}{2} \times (1.5) t^2 \quad \dots\dots\dots(1)$$

$$\text{and } (x + 150) = \frac{1}{2} \times (2) t^2 \quad \dots\dots\dots(2)$$

Here t is the common time.

From eqs. (1) and (2) we have

$$= \frac{x+150}{x} = \frac{2}{1.5}$$

Solving we get $x = 450$ metre (truck) and

$x + 150 = 600$ metre (car).

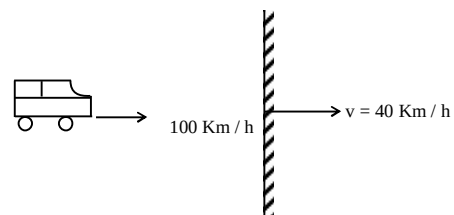
Substituting the value of x in eq. (1),

we get $450 = \frac{1}{2} (1.5) t^2$

$$\therefore t = \sqrt{\frac{450 \times 2}{1.5}} = \sqrt{600} = 24.5 \text{ sec.}$$

Hence correct answer is (C)

7. The back side of the truck which is moving with 40 km/h has shining surface. The surface can be treated as a plane mirror. A motorist trying to overtake the truck from behind is moving with speed of 100 km/h. Find the velocity of the image of motorist observed to himself and to a person standing on the roadside.



- (A) 10 kmph
(B) 30 kmph
(C) 20 kmph
(D) 5 kmph

Ans. C

Sol. Velocity of object = 100 km/h.

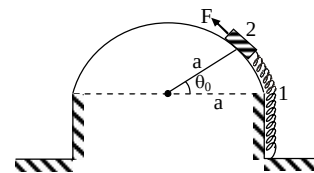
Velocity of mirror = 40 km/h

Due to velocity of mirror the image will move towards right with double the velocity i.e., 80 km/h. and due to the motion of the object the image will move with velocity 100 km/h towards left. Hence net velocity of the image is $100 - 80 = 20$ kmph towards left.

Velocity of image observed to the motorist = $20 + 100 = 120$ kmph towards left.

Velocity of image observed to the standing person = 20 kmph.

8. In the position 1 (figure) the spring of constant K is undeformed. Find the work done by the force $\vec{F}$ (which is always directed along the tangent to the smooth hemi spherical surface) on the small block of mass m to shift it from the position 1 to position 2 slowly.



- (A) $mg R \sin \theta_0 + \frac{KR^2\theta_0^2}{2}$
(B) $mg R \sin \theta_0$
(C) $mg R \sin \theta_0 + \frac{KR^2\theta_0^2}{4}$
(D) Zero

Ans. A

Sol. Let us locate the block at an arbitrary angular position $\theta < \theta_0$

$$dW = \vec{F} \cdot d\vec{r} = Fds = F(Rd\theta) \quad \dots(1)$$

Now from the condition of equilibrium of the block, we have

$$F = mg \cos \theta + KR \theta \quad \dots(2)$$

from eq. (1) & (2)

$$dW = mg R \cos \theta d\theta + KR^2 \theta d\theta$$

Hence the sought work

$$W = \int dW = \int_0^{\theta_0} \cos \theta d\theta + KR^2 \int_0^{\theta_0} \theta d\theta$$

$$W = mg R \sin \theta_0 + \frac{KR^2 \theta_0^2}{2}$$

9. The ionization potential of Li^{++} ion is 122.4 V. Find the value of its first excitation potential.

- (A) 91.8v
(B) 45.9 V
(C) 122.4V
(D) 28.4 V

Ans. A

Sol. According to given data, ionization energy of Li^{++} ion is 122.4 eV

$\therefore$ Excitation energy of this ion

$$= 122.4 \left[\frac{1}{1^2} - \frac{1}{2^2} \right]$$

$$= 122.4 \left[1 - \frac{1}{4} \right]$$

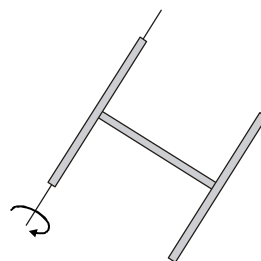
$$= 122.4 \times \frac{3}{4} \text{ eV}$$

$$\therefore E_{\text{ex}} = 91.8 \text{ eV}$$

Hence, the value of the first excitation potential
= 91.8 V

10. A rigid body is made of three identical thin rods, each of length L fastened together in the form of the letter H. The body is free to rotate about a horizontal axis that runs along the length of one of the legs of the H. The body is allowed to fall from rest from a position in which the plane of the H is horizontal. What is the angular speed of the body

when the plane of H is vertical $\frac{3}{2} \sqrt{\frac{g}{L}}$



(A) $\frac{5}{2} \sqrt{\frac{g}{L}}$

(B) $\frac{3}{2} \sqrt{\frac{g}{L}}$

(C) $\frac{3}{4} \sqrt{\frac{g}{L}}$

(D) $\sqrt{\frac{g}{L}}$

Ans. B

Sol. Moment of inertia of the system about the given axis

$$I = I_A + I_B + I_C$$

Now as rod is thin

$$I_A = \Sigma m \times 0^2 = 0$$

Rod B is rotating about one end

$$I_B = ML^2 / 3$$

And for rod C all points are always at distance L from the axis of rotation, so

$$I_C = \Sigma mL^2 = ML^2$$

$$\therefore I = 0 + (ML^2 / 3) + ML^2 = 4ML^2 / 3$$

So if ω is the desired angular speed, gain in kinetic energy due to rotation of H from horizontal to vertical position.

$$K_R = \frac{1}{2} I \omega^2 = \frac{1}{2} \left[\frac{4}{3} ML^2 \right] \omega^2$$

And loss in potential energy of the system in doing so

$$= 0 + Mg \frac{L}{2} + MgL = \frac{3}{2} MgL$$

So by conservation of mechanical energy

$$(2 / 3) ML^2 \omega^2 = (3/2) MgL$$

$$\text{or } \omega = \frac{3}{2} \sqrt{\frac{g}{L}}$$

11. A radioactive sample has 6.0×10^{18} active nuclei at a certain instant. How many of these nuclei will still be in the same active state after two half-lives?

(A) 3×10^{18}

(B) 1.5×10^{18}

(C) 0.75×10^{18}

(D) 6×10^{18}

Ans. B

Sol. In one half-life the number of active nuclei reduces to half the original number. Thus, in two half-

lives the number is reduced to $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)$ of the original number. The number of remaining active

nuclei is, therefore, $6.0 \times 10^{18} \times \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) = 1.5 \times 10^{18}$

12. One solid sphere, having mass 1kg & diameter 0.3m is suspended from a wire. If the twisting couple per unit twist for the wire is 6×10^{-3} N-m/radian, then the time period of small oscillations will be:

- (A) 0.7 sec
(B) 7.7 sec
(C) 77 sec
(D) 777 sec

Ans. B

Sol. $T = 2\pi \sqrt{\frac{I}{C}}$ (1)

$I = \frac{2}{5} MR^2$ (2)

By equation (1) & (2)

$T = 2\pi \sqrt{\frac{2MR^2}{5C}}$

given, $M = 1\text{kg}$

$R = 0.15\text{ m}$

$C = 6 \times 10^{-3}\text{ N-m/rad.}$

$T = ?$

$\Rightarrow T = 6.28 \sqrt{\frac{0.4 \times 1 \times (0.15)^2}{6 \times 10^{-3}}}$

$T = 6.28 \sqrt{\frac{9 \times 10^{-3}}{6 \times 10^{-3}}} = 6.28 \times \sqrt{1.5}$

$T = 7.7\text{ sec.}$

13. Find the constant b in Moseley's equation $\sqrt{\nu} = a(Z - b)$ from the following data.

Element	Z	Wavelength of K_{α} X-ray
Mo	42	71 pm
Co	27	178.5 pm

- (A) 1.43
(B) 1.11
(C) 1
(D) 1.37

Ans. D

Sol. Moseley's equation is

$\sqrt{\nu} = a(Z - b).$

$$\text{Thus } \sqrt{\frac{c}{\lambda_1}} = a(Z_1 - b) \quad \dots(i)$$

$$\text{and } \sqrt{\frac{c}{\lambda_2}} = a(Z_2 - b) \quad \dots(ii)$$

Using equations (i) and (ii), we get

$$\sqrt{c} \left(\frac{1}{\sqrt{\lambda_1}} - \frac{1}{\sqrt{\lambda_2}} \right) = a(Z_1 - Z_2)$$

$$\Rightarrow a = \frac{\sqrt{c}}{(Z_1 - Z_2)} \left(\frac{1}{\sqrt{\lambda_1}} - \frac{1}{\sqrt{\lambda_2}} \right)$$

Putting given values

$$a = 5.06 \times 10^7 \text{ Hz}$$

Dividing equation (i) by (ii), we get

$$\frac{\sqrt{\lambda_2}}{\sqrt{\lambda_1}} = \frac{Z_1 - b}{Z_2 - b}$$

$$\Rightarrow \sqrt{\frac{178.5}{71}} = \frac{42 - b}{27 - b}$$

$$\Rightarrow b = 1.37$$

14. A 70 cm long sonometer wire is in unison with a tuning fork. If the length of the wire is decreased by 1.0 cm, it produces 4 beats per sec. with the same tuning fork. Find the frequency of the tuning fork -

- (A) 276 sec^{-1}
 (B) 272 sec^{-1}
 (C) 269 sec^{-1}
 (D) 280 sec^{-1}

Ans. A

Sol. Let the frequency of the fork be n . In the first case, the length of the wire is 70 cm (0.70 m). Therefore,

$$n = \frac{1}{2 \times 0.70} \sqrt{\frac{T}{m}} \quad \dots(1)$$

On decreasing the length of the wire, its frequency will increase. Hence in the second case when the length is 69 cm = 0.69 m, then

$$n + 4 = \frac{1}{2 \times 0.69} \sqrt{\frac{T}{m}} \quad \dots(2)$$

dividing equation (1) by (2), we get

$$\Rightarrow \frac{n}{n+4} = \frac{0.69}{0.70}$$

$$\Rightarrow n \times 0.70 = (n + 4) \times 0.69$$

$$\Rightarrow n = \frac{4 \times 0.69}{0.01} = 276 \text{ sec}^{-1}$$

15. Mr. Sanjay wants to calculate the current and power dissipated in an LCR series circuit. He connected 100W resistance to an AC source of 200 V and angular frequency 300 radian/sec. When he removed only the capacitance, the current was found to be lagging behind the voltage by 60° . While on removing the inductance he found, the current leading the voltage by 60° . Find the value of current and the power dissipated obtained by him

- (A) 200 Watt
(B) 100 Watt
(C) 400 Watt
(D) 300 Watt

Ans. A

Sol. $\tan 60^\circ = \frac{\omega L}{R}$, $\tan 60^\circ = \frac{1/\omega C}{R}$.

$$\therefore \omega L = \frac{1}{\omega C}.$$

$\therefore$ impedance of the circuit,

$$Z = \left[R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2 \right]^{\frac{1}{2}} = R.$$

Current in the circuit,

$$i_0 = \frac{V_0}{Z} = \frac{V_0}{R} = \frac{200}{100} = 2 \text{ ampere.}$$

$$\text{Average power, } \bar{P} = \frac{1}{2} V_0 i_0 \cos \phi.$$

$$\text{But } \tan \phi = \frac{\omega L - 1/\omega C}{R} = 0, \quad \therefore \cos \phi = 1.$$

$$\therefore \bar{P} = \frac{1}{2} \times 200 \times 2 \times 1 = 200$$

16. The length of a wire between the two ends of a sonometer is 105 cm. Where the two bridges should be placed so that the fundamental frequencies of the three segments are in the ratio of 1:3:15.

- (A) 75 cm, 25 cm,
(B) 25 cm, 75 cm,
(C) 75 cm, 100 cm,
(D) None of these

Ans. C

Sol. From the law of length of stretched string, we have $n_1 \ell_1 = n_2 \ell_2 = n_3 \ell_3$

Here $n_1 : n_2 : n_3 = 1 : 3 : 15$

$$\therefore \frac{\ell_1}{\ell_2} = \frac{n_1}{n_2} = \frac{3}{1} \text{ and } \frac{\ell_1}{\ell_3} = \frac{n_1}{n_3} = \frac{15}{1}$$

$$\ell_2 = \frac{\ell_1}{3} \text{ and } \ell_3 = \frac{\ell_1}{15}$$

The total length of the wire is 105 cm.

Therefore $\ell_1 + \ell_2 + \ell_3 = 105$

$$\text{or } \ell_1 + \frac{\ell_1}{3} + \frac{\ell_1}{15} = 105 \text{ or } \frac{21\ell_1}{15} = 105$$

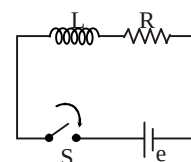
$$\ell_1 = \frac{105 \times 15}{21} = 75 \text{ cm.}$$

$$\therefore \ell_2 = \frac{\ell_1}{3} = \frac{75}{3} = 25 \text{ cm}$$

$$\ell_3 = \frac{\ell_1}{15} = \frac{75}{15} = 5 \text{ cm}$$

Hence the bridge should be placed at 75 cm and $(75 + 25) = 100$ cm from one end.

17. In an LR circuit as shown in fig. when the switch is closed how much time will it take the current to grow a value $\frac{\eta \epsilon}{R}$ (where $\eta < 1$)



(A) $(L/R) \ln \left| \frac{1}{\eta} \right|$

(B) $(L/R) \ln \left| \frac{1}{1-\eta} \right|$

(C) $(2L/R) \ln \left| \frac{1}{\eta} \right|$

(D) None of these

Ans. B

Sol. We know that

$$i = \frac{\epsilon}{R} (1 - e^{-t/\tau})$$

$$i = \eta \frac{\epsilon}{R} \text{ (given)}$$

$$\eta \frac{\epsilon}{R} = \frac{\epsilon}{R} (1 - e^{-t/\tau})$$

$$\text{or } e^{-t/\tau} = 1 - \eta$$

$$\text{or } t = \tau \ln \left| \frac{1}{1-\eta} \right|$$

18. An infinite number of charges each equal to q are placed along the x -axis at $x = 1, x = 2, x = 4, x = 8, \dots$ and so on. Find the electric field at the point $x = 0$ due to this set of charges.

(A) $\frac{1}{4\pi\epsilon_0} \left[\frac{2}{3} q \right]$

(B) $\frac{1}{4\pi\epsilon_0} \left[\frac{8}{3} q \right]$

(C) $\frac{1}{4\pi\epsilon_0} \left[\frac{4}{3} q \right]$

(D) $\frac{1}{4\pi\epsilon_0} \left[\frac{5}{3} q \right]$

Ans. C

Sol. As potential V and intensity E due to a point charge at position x are respectively given by-

$$\begin{array}{ccccccc} & q & q & q & q & q & \\ & \bullet & \bullet & \bullet & \bullet & \bullet & \\ x=0 & 1 & 2 & 4 & 8 & 16 & \end{array}$$

$$E = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{1^2} + \frac{q}{2^2} + \frac{q}{4^2} + \dots \right]$$

$$= \frac{q}{4\pi\epsilon_0} \left[1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \right]$$

i.e.,

$$E = \frac{q}{4\pi\epsilon_0} \frac{1}{[1 - (1/4)]} = \frac{1}{4\pi\epsilon_0} \left[\frac{4}{3} q \right]$$

19. A large plane charge sheet having surface charge density $\sigma = 2.0 \times 10^{-6} \text{ C/m}^2$ lies in the X - Y plane. Find the flux of the electric field through a circular area of radius 1 cm lying completely in the region where x, y, z are all positive and with its normal making an angle of 60° with the Z -axis.

(A) $15 \text{ N} \cdot \text{m}^2 / \text{C}$

(B) $12.5 \text{ N} \cdot \text{m}^2 / \text{C}$

(C) $17.5 \text{ N} \cdot \text{m}^2 / \text{C}$

(D) None of these

Ans. C

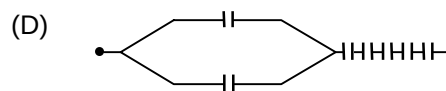
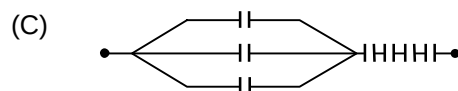
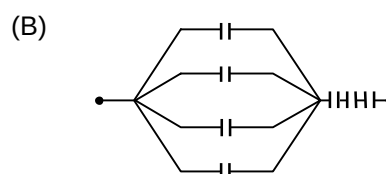
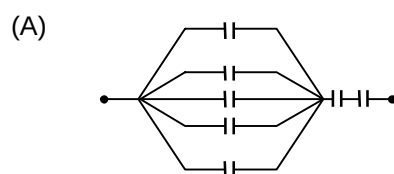
Sol. The electric field near the plane charge sheet is $E = \sigma/2\epsilon_0$ in the direction away from the sheet. At the given area, the field is along the Z-axis.

$$\begin{aligned} \text{The area} &= \pi r^2 = 3.14 \times 1 \text{ cm}^2 \\ &= 3.14 \times 10^{-4} \text{ m}^2. \end{aligned}$$

The angle between the normal to the area and the field is 60° .

$$\begin{aligned} \text{Hence, the flux} &= \vec{E} \cdot \Delta \vec{S} = E \Delta S \cos \theta \\ &= \frac{\sigma}{2\epsilon_0} \pi r^2 \cos 60^\circ \\ &= \frac{2.0 \times 10^{-6} \text{ C/m}^2}{2 \times 8.85 \times 10^{-12} \text{ C}^2/\text{N-m}^2} \times \\ &\quad (3.14 \times 10^{-4} \text{ m}^2) \frac{1}{2} \\ &= 17.5 \text{ N-m}^2/\text{C} \end{aligned}$$

20. Seven capacitors each of capacitance 2 mF are to be connected in a configuration to obtain an effective capacitance of $(10/11)$ mF. Which of the combination(s) shown in figure will achieve the desired result?

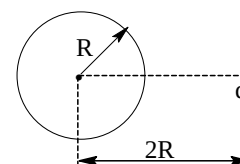


Ans. A

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A point charge is placed outside a spherical shell as shown in the figure. Find the electric potential (in kilo volt) on the surface of shell (Take: $q=1\mu\text{C}$ and $R=50\text{cm}$)

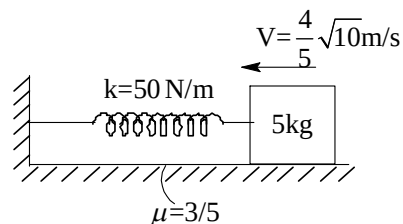


Ans. 00009.00

Sol. Field inside the shell is zero, so potential is uniform

$$V = \frac{kq}{2R} = 9 \text{ kilo volts}$$

22. A spring block system is placed on a horizontal rough surface as shown in the figure. The block is given velocity when the spring is in natural length. The total distance travelled by the block before it finally comes to rest is $\frac{2}{k}$ meters. Find the value of k



Ans. 00005.00

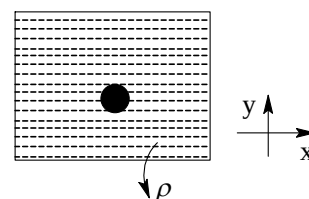
Sol.
$$\frac{-3}{5}(50)x - \left(\frac{1}{2} \times 50x^2\right) = 0 - \frac{1}{2} \times 5 \left(\frac{4}{5}\sqrt{10}\right)^2$$

$$25x^2 + 30x - 16 = 0 \Rightarrow x = \frac{2}{5} \text{ meters}$$

So at maximum compression spring force $= 50\left(\frac{2}{5}\right) = 20\text{N}$ and friction force $= \frac{3}{5}(50) = 30\text{N}$.

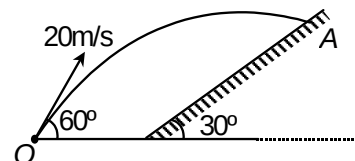
Hence the block will not move so distance $= 2/5\text{m}$.

23. A small spherical ball of density same as that of liquid is released from rest in a vessel filled completely with a liquid and accelerating with acceleration $5\hat{i} + 5\hat{j} \text{ m/s}^2$ as shown in the figure. Find the initial acceleration of ball just after the release with respect to vessel.



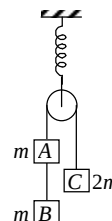
Ans. 00000.00

24. A ball is projected from the point O with velocity 20 m/s at an angle of 60° with horizontal as shown in figure. At highest point of its trajectory it strikes a smooth plane of inclination 30° at point A. The collision is perfectly inelastic. The maximum height from the ground attained by the ball is $\frac{75}{k}$ meter. Find the value of k ? ($g = 10 \text{ m/s}^2$)



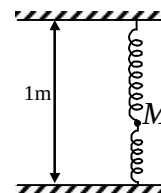
Ans. 00004.00

25. In the figure shown, pulley and spring are ideal and strings are light and inextensible. Initially all the bodies are at rest when string connecting A and B is cut. Find initial acceleration of C.



Ans. 00000.00

26. A small ball of mass $M=1\text{kg}$ is attached to two identical springs of constant 100N/m , which are attached to floor and roof. The springs are initially unstretched. By what distance ball comes down to attain equilibrium? (in cm)



Ans. 00005.00

27. In YDSE, when a glass plate of refractive index 1.5 of thickness t is placed in the path of one of the interfering beams of wavelength λ , intensity at the position where central maximum occurred previously remain unchanged. If the minimum thickness of the glass plate is $k\lambda$. Find the value of k

Ans. 00002.00

Sol. $t(\mu - 1) = \lambda \Rightarrow t(1.5 - 1) = \lambda \Rightarrow t = 2\lambda$

28. If the ionization potential of hydrogen atom is 13.6eV , the energy required to remove from the third orbit of hydrogen atom is $k/2\text{eV}$. Find the value of k

Ans. 00003.00

Sol. Energy required removing the electron from the third orbit

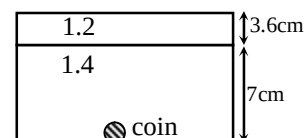
$$= \frac{13.6}{3^2} \text{eV} = 1.5 \text{eV} = 1.5 \times 1.6 \times 10^{-19} \text{J} = 3 \text{eV}$$

29. When an object is at a distance 3meters and 1meters from the optical centre of a convex lens, a real and virtual images are formed respectively, with the same magnification. Find the focal length of the lens (in meters)

Ans. 00002.00

Sol. $\frac{f}{-u_1 + f} = -\frac{f}{(-u_2 + f)} \Rightarrow -u_2 + f = u_1 - f \Rightarrow f = \frac{u_1 + u_2}{2} = 2$

30. A man observes a coin placed at the bottom of a beaker which contains two immiscible liquids of refractive indices 1.2 and 1.4 as shown in the figure. Find the depth of the coin below the surface, as observed from above.



Ans. 00008.00

Sol. $d = \frac{3.6}{1.2} + \frac{7}{1.4} = 8$

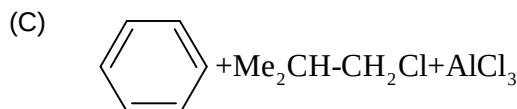
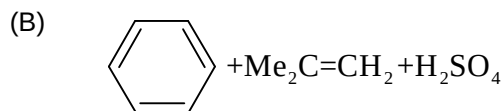
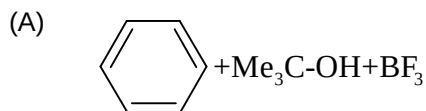
Chemistry

PART – B

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

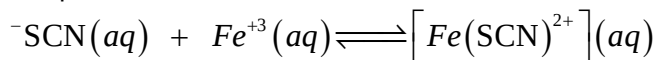
31. In which of the following reactions tertiary butyl benzene is formed?



(D) All of these

Ans. D

32. In equilibrium:

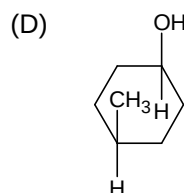
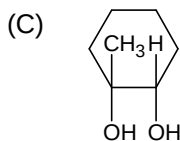
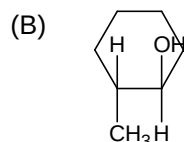
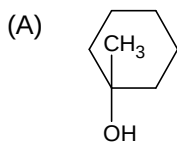
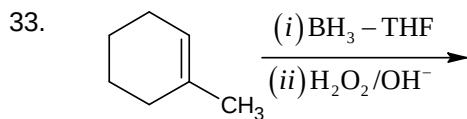


colourless yellow

If thiocyanate ions are added in equilibrium mixture

- (A) The solution becomes colourless
 (B) The yellow colour of solution deepens.
 (C) The red colour of the solution deepens.
 (D) Concentration of $[\text{Fe}(\text{SCN})]^{2+}$ ion will decrease.

Ans. C



Ans. B

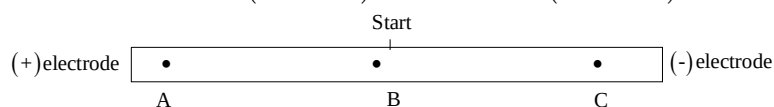
Sol. Anti-markownikoff addition of H_2O and syn-mechanism

34. Which of the following is incorrect?

- (A) The heats of hydration of the dipositive alkaline earth metal ions decrease with an increase in their ionic size
- (B) $NaNO_3$ forms Na_2O_2 on heating
- (C) Hydration of alkali metal ion is less than that of IIA group
- (D) Alkaline earth metal ions, because of their much larger charge to size ratio, exert a much stronger electrostatic attraction on the oxygen of water molecule surrounding them.

Ans. B

35. Referring to figure, identify the three spots obtained by electrophoresis at $p^H = 7$ of a mixture of the proteins pepsin ($p^I = 1.1$) haemoglobin ($p^I = 6.8$) and Lysozyme ($p^I = 11.0$):

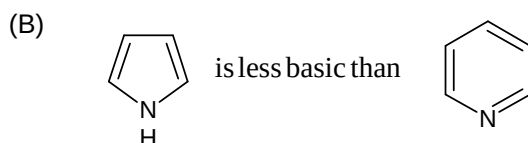
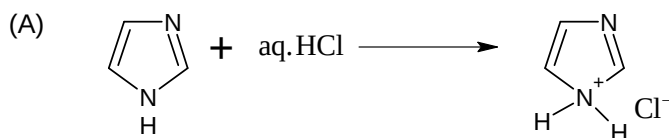


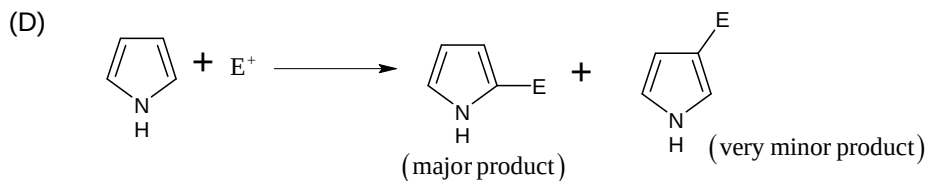
Which of the following options are correct about position of various proteins?

- (A) Lysozyme will be present at spot A
- (B) pepsin will be present at spot B
- (C) pepsin will be present at spot A
- (D) haemoglobin will be present at spot C

Ans. C

36. Which of the following is incorrect statement/reaction?





Ans. A

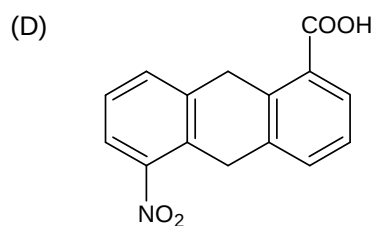
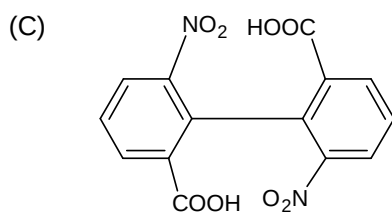
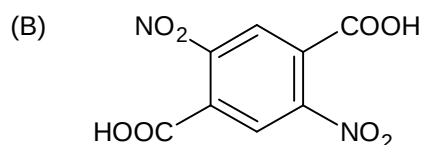
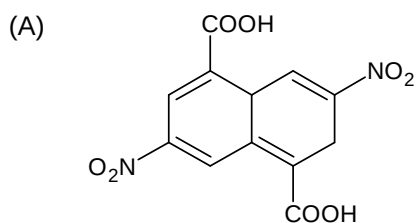
37. Which of the following statement about diffusion of gas is true?

- (A) Rate of diffusion of a gas will increase if an inert gas is added to gaseous mixture and allowed to diffuse through a pin hole
- (B) Rate of effusion of O_2 is $\sqrt{2}$ times that of SO_2 , for every composition of O_2 and SO_2 mixture
- (C) Rate of effusion decrease on increasing the temperature
- (D) All are true

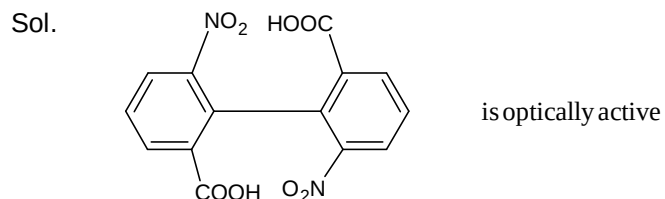
Ans. C

Sol. Rate of effusion $\propto \frac{1}{\sqrt{T}}$

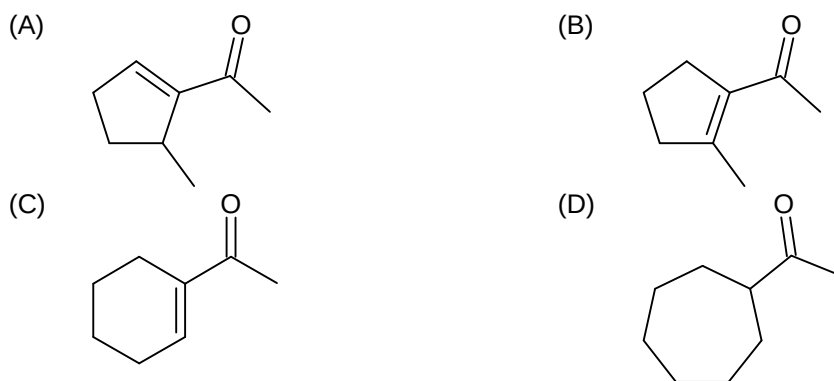
38. Which of the following carboxylic acid could be resolved by reaction with an enantiomerically pure chiral amine?



Ans. C



39. 2, 7-Octanedione on treatment with aq. KOH and 100°C produces?



Ans. B

40. The compound which gives Carbylamine reaction.



Ans. A

Sol. Primary amines give Carbylamine test.

41. Which of the following combinations are correct for a binary solution, in which the solute and solvent are liquid?

- (A) C_6H_6 and $\text{C}_6\text{H}_5\text{CH}_3$; $\Delta H_{\text{sol}} > \Delta V_{\text{sol}} = 0$
- (B) $\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ and CHCl_3 ; $\Delta H_{\text{sol}} < 0$; $\Delta V_{\text{sol}} < 0$
- (C) H_2O and HCl ; $\Delta H_{\text{sol}} > 0$; $\Delta V_{\text{sol}} < 0$
- (D) H_2O and $\text{C}_2\text{H}_5\text{OH}$; $\Delta H_{\text{sol}} < 0$; $\Delta V_{\text{sol}} > 0$

Ans. B

42. Crystal field splitting energy (Δ_0) of complexes $[CrCl_6]^{3-}$, $[Cr(CN)_6]^{3-}$ and $[Cr(NH_3)_6]^{3+}$ are x, y and $z \text{ cm}^{-1}$ respectively. Then correct order Δ_0 values is:

- (A) $x < y < z$
 (B) $x < z < y$
 (C) $x > y > z$
 (D) $x > z > y$

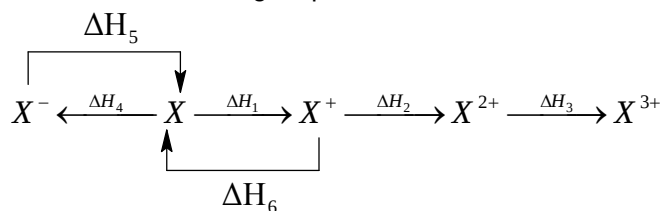
Ans. B

43. Optical isomerism is not shown by the complex:

- (A) $[Cr(ox)_3]^{3-}$
 (B) $Cis[Co(en)_2Cl_2]^+$
 (C) $Trans[Co(en)_2Cl_2]^+$
 (D) $[Cr(en)_3]^{3+}$

Ans. C

44. Consider the following sequence of reaction:



If electronic configuration of element X is $[Ne]3s^1$, then which of the following order is incorrect regarding given enthalpies?

- (A) $|\Delta H_4| = |\Delta H_5|$
 (B) $|\Delta H_2| > |\Delta H_1|$
 (C) $|\Delta H_2| > |\Delta H_3|$
 (D) $|\Delta H_1| = |\Delta H_6|$

Ans. C

Sol. When a reaction is reversed, magnitude of enthalpy change remains same.

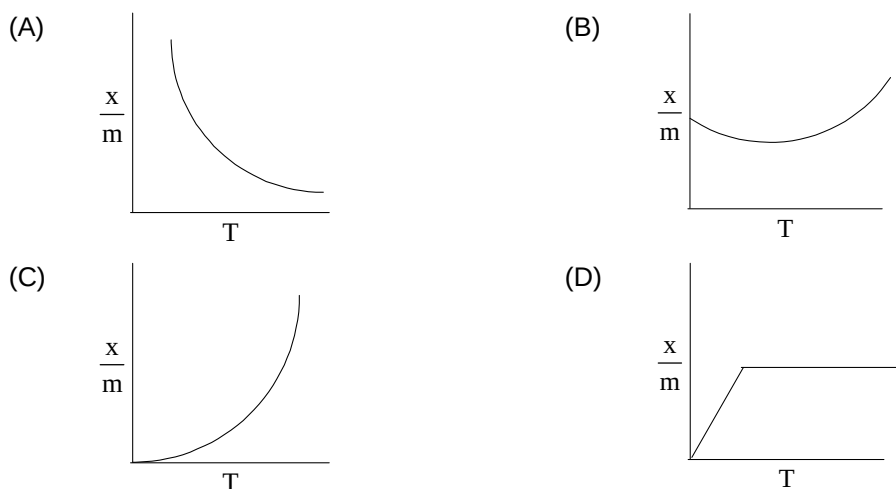
45. Which of the following does not represent the correct sequence of property indicated?

- (A) $\text{Sc}^{3+} > \text{Cr}^{3+} > \text{Fe}^{3+} > \text{Mn}^{3+}$ - Stability of +3 ions
 (B) $\text{Sc} < \text{Ti} < \text{Cr} < \text{Mn}$ – number of oxidation states
 (C) $\text{Mn}^{2+} < \text{Ni}^{2+} < \text{Co}^{2+} < \text{Fe}^{2+}$ - Magnetic moment
 (D) $\text{FeO} > \text{CoO} > \text{NiO} > \text{CuO}$ – Basic nature

Ans. C

Sol. Magnetic moment sequence is $\text{Mn}^{2+} (3d^5) > \text{Fe}^{2+} (3d^6) > \text{Co}^{2+} (3d^7) > \text{Ni}^{2+} (3d^8)$

46. Which plot is the adsorption isobar for chemisorption?



Ans. C

Sol. Chemisorption increases with increase in temperature initially and then decreases.

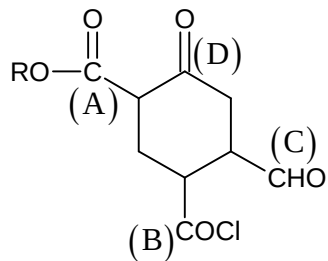
47. Which of the following reaction is non-redox?

- (A) $2\text{NaNO}_3 \longrightarrow 2\text{NaNO}_2 + \text{O}_2$
 (B) $\text{KCl} + \text{NaNO}_3 \longrightarrow \text{KNO}_3 + \text{NaCl}$
 (C) $\text{Fe} + \text{H}_2\text{SO}_4 \longrightarrow \text{FeSO}_4 + \text{H}_2$
 (D) $4\text{Ag} + 8\text{CN}^- + \text{O}_2 + \text{H}_2\text{O} \longrightarrow 4[\text{Ag}(\text{CN})_2]^- + 4\text{OH}^-$

Ans. B

Sol. Oxidation state of all species remain unchanged so it is not redox

48. Find the correct order of reactivity of carbonyl groups with LiAlH_4 :

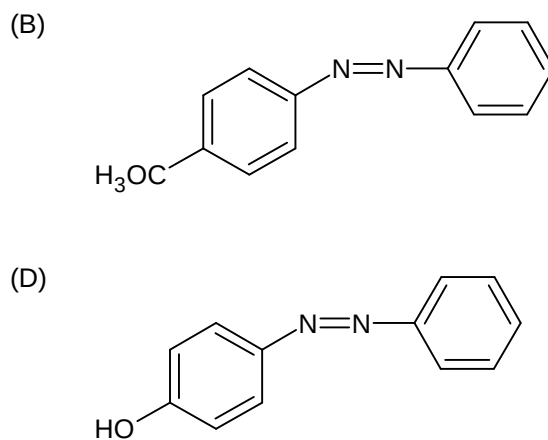
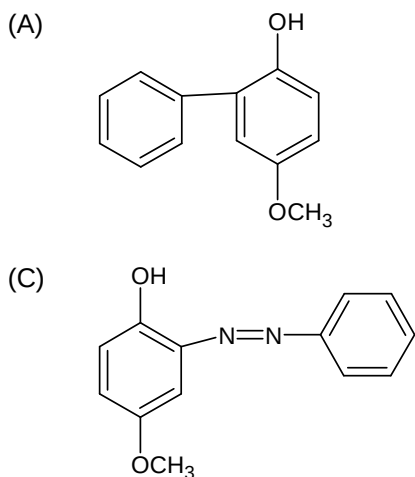
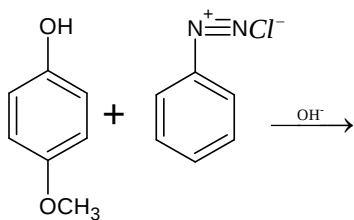


- (A) $A > B > C > D$
 (B) $B > C > D > A$
 (C) $D > C > B > A$
 (D) $B > D > C > A$

Ans. B

Sol. Reactivity of carbonyl compounds follow the order acid chlorides > aldehydes > ketones > ester

49. Product of reaction is:



Ans. C

Sol. OH^- Converts phenol to phenoxide ion which is strongly activating so attack occurs at ortho position w.r.t. hydroxyl group

50. A sulphide of a metal (M) with formula MS is white coloured ppt which is soluble in acidic solution ($\text{pH} < 6$). Metal cation M is:

(A) Pb^{2+}

(B) Zn^{2+}

(C) Hg^{2+}

(D) Cu^{2+}

Ans. B

Sol. ZnS is white coloured ppt which is soluble in acidic solution below $\text{pH} = 6$

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. Calculate value of $\frac{X+Y+Z}{10}$, here X is $\text{O}-\text{N}-\text{O}$ bond angle in NO_3^- Y is $\text{O}-\text{N}-\text{O}$ bond angle in NO_2^+ and Z is $\text{F}-\text{Xe}-\text{F}$ adjacent bond angle in XeF_4 .

Ans. 00039.00

Sol. $X=120$ (sp^2 -hybridization)
 $Y=180$ (sp -hybridization)
 $Z=90$ (sp^3d^2 -hybridization)

$$\frac{120+180+90}{10} = \frac{390}{10} = 39$$

52. The overall charge present on the cyclic silicate anion $[\text{Si}_6\text{O}_{18}]^{n-}$ is ____

Ans. 00012.00

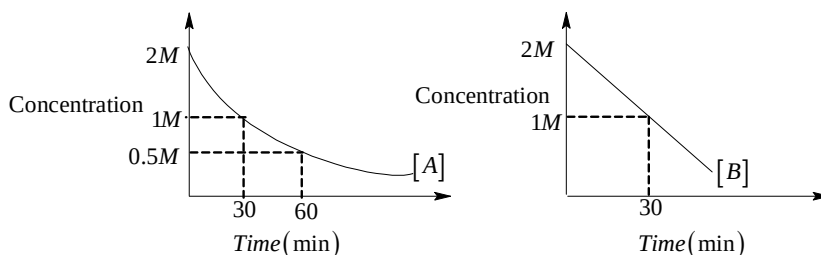
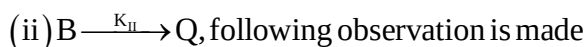
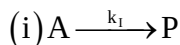
Sol. Given silicate anion is $[(\text{SiO}_3)^{2-}]_6$
 $\therefore n = 12$

53. How many of the following molecules or ions have sp^3d hybridisation of their central atom ?
 I_3^- , SbF_6^- , F_3^+ , ClO_3^- , XeO_2F_2 , XeO_6^{4-} , SF_4

Ans. 00003.00

Sol. I_3^- , XeO_2F_2 , SF_4

54. For the reactions



Calculate $\frac{k_I}{K_{II}}$, where k_I and k_{II} are rate constant for the respective reaction:

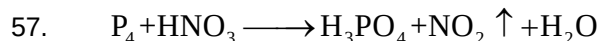
Ans. 00000.69

55. One mole of a real gas is subjected to a process from (2 bar, 30 L, 300 K) to (2 bar, 40 L, 500 K). The molar heat capacity of gas at constant pressure is 40 J/K mol. What is the change in internal energy (in kJ) of the gas in this process?

Ans. 00006.00

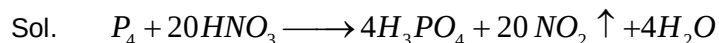
56. A saturated solution of silver benzoate, $AgOCOC_6H_5$, has pH of 8.6 K_a for benzoic acid is 5.0×10^{-5} . The value of K_{sp} for silver benzoate is $x \times 10^{-3}$. Then the value of x is? ($\log 2 = 0.3$)

Ans. 00006.40



Product mixture is titrated with 0.1M $NaOH$, indicator shows colour change at second end point and 100ml of $NaOH$ is used for that. How much volume (in mL) of 1M HNO_3 is used in above reaction?

Ans. 00025.00



$$m.\text{moles of } NaOH = 2 \times m.\text{moles of } H_3PO_4 \Rightarrow m.\text{moles } H_3PO_4 = \frac{100 \times 0.1}{2} = 5$$

$$m.\text{moles of } HNO_3 = \frac{20}{4} \times 5 = 25$$

$$\text{Volume of } HNO_3 = 25 \text{ ml}$$

58. Potassium crystallizes in *BCC* lattice with a unit cell length $a = 5.0 \text{ \AA}$. What is the distance (in $\text{\AA}$) between next nearest neighbours?

Ans. 00005.00

59. The dissociation constant of a weak acid is 1.6×10^{-5} and molar conductivity at infinite dilution is $380 \times 10^{-4} \text{ Sm}^2 \text{mol}^{-1}$. If the cell constant is 0.01 m^{-1} the conductance (in S) of 0.01 M acid solution is _____

Ans. 00001.52

Sol. $\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{1.6 \times 10^{-5}}{0.01}}$
 $\lambda_m = \alpha \lambda_m^\circ = \frac{K \times 1000}{\text{molarity}}$ when λ_m is in $\text{Scm}^2 \text{mol}^{-1}$
 $G = \frac{K}{\text{cell constant}}$

60. For 3s orbital of hydrogen atom, the normalised wave function is

$$\psi_{3s} = \frac{1}{81\sqrt{3\pi}} \left(\frac{1}{a_0} \right)^{3/2} \left[27 - \frac{18r}{a_0} + \frac{2r^2}{a_0^2} \right] e^{\frac{-r}{3a_0}}$$

If distance between the radial nodes is d , calculate the value of $\frac{d}{1.73a_0}$

Ans. 00003.00

Mathematics**PART – III****SECTION – A**
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. $\int \frac{3x^2 + 4x - 1}{(x^2 + 1)^2 \sqrt{x+1}} dx$ is equal to:

(A) $\frac{\sqrt{x+1}}{x^2 + 1} + c$

(B) $-\frac{2\sqrt{x+1}}{x^2 + 1} + c$

(C) $\frac{2}{(x^2 + 1)\sqrt{x+1}}$

(D) $\frac{-x}{(x^2 + 1)\sqrt{x+1}} + c$

Ans. B

Sol.
$$\begin{aligned} \int \frac{3x^2 + 4x - 1}{(x^2 + 1)^2 \sqrt{x+1}} dx &= \int \frac{4x^2 + 4x - x^2 - 1}{(x^2 + 1)^2 \sqrt{x+1}} dx \\ &= 4 \int \frac{x(x+1) dx}{(x^2 + 1)^2 \sqrt{x+1}} - \int \frac{dx}{(x^2 + 1)\sqrt{x+1}} \\ &= 4 \int \frac{x(x+1) dx}{(x^2 + 1)^2 \sqrt{x+1}} \left[\frac{2\sqrt{x+1}}{x^2 + 1} \int \frac{-1}{(x^2 + 1)^2} \cdot 2x \cdot 2\sqrt{x+1} \cdot dx \right] \\ &= 4 \int \frac{x(x+1) dx}{(x^2 + 1)^2 \sqrt{x+1}} - \frac{2\sqrt{x+1}}{x^2 + 1} \\ &= \frac{-2\sqrt{x+1}}{x^2 + 1} + c \end{aligned}$$

62. The general solution of the differential $y\left(xy - (x^2 - y^2)^2\right)dx = \left\{y^3 - x(x^2 - y^2)^2\right\}dy$ is:

- (A) $\frac{y}{x} + \frac{1}{x^2 - y^2} = c$
- (B) $\frac{x}{y} - \frac{1}{2} \ln \left| \frac{x-y}{x+y} \right| = c$
- (C) $\frac{2x}{y} + \frac{1}{x^2 - y^2} = c$
- (D) $\frac{2x}{y} - \frac{1}{x^2 - y^2} = c$

Ans. C

Sol. $y\left(xy - (x^2 - y^2)^2\right).dx = \left\{y^3 - x(x^2 - y^2)^2\right\}dy$
 $\Rightarrow xy^2dx - y(x^2 - y^2)^2dx = y^3dy - x(x^2 - y^2)^2dy$
 $\Rightarrow y^2(xdx - ydy) = (x^2 - y^2)^2(ydx - xdy)$
 $\Rightarrow xdx - ydy = (x^2 - y^2)^2 \frac{(ydx - xdy)}{y^2}$
 $\frac{2(xdx - ydy)}{(x^2 - y^2)^2} = \frac{2(ydx - xdy)}{y^2}$
 $\int \frac{d(x^2 - y^2)}{(x^2 - y^2)^2} = 2 \int d\left(\frac{x}{y}\right)$
 $\frac{-1}{x^2 - y^2} = \frac{2x}{y} + C$
 $\therefore \frac{2x}{y} + \frac{1}{x^2 - y^2} = C$

63. In how many ways can the letters of the word ARRANGE be arranged so that the two A's are always together and two R's are never together.

- (A) 240
- (B) 260
- (C) 300
- (D) 340

Ans. A

64. The equation of a circle whose diameter is the common chord of the circle $x^2 + y^2 - 4x - 8y - 5 = 0$ and $x^2 + y^2 - 12x + 2y + 29 = 0$ is:

(A) $41x^2 + 41y^2 - 396x - 38y + 781 = 0$

(B) $2x^2 + 2y^2 - 16x - 6y + 25 = 0$

(C) $41x^2 + 41y^2 + 396x - 38y - 781 = 0$

(D) $17x^2 + 17y^2 - 300x + 154y + 901 = 0$

Ans. A

Sol. $C : (x^2 + y^2 - 4x - 8y - 5) + \lambda (x^2 + y^2 - 12x + 2y + 29) = 0$

Centre : $\left(\frac{2+6\lambda}{1+\lambda}, \frac{4-\lambda}{1+\lambda} \right)$

Common chord $4x - 5y - 17 = 0$

Centre lies on common chord

$\therefore \lambda = \frac{29}{12}$

$\therefore \text{centre} \left(\frac{1+6(29/12)}{1+29/12}, \frac{4-29/12}{1+29/12} \right)$

$\therefore \text{centre} \left(\frac{198}{41}, \frac{19}{41} \right)$

65. $\int_0^{\frac{11\pi}{3}} \sqrt{1 + \cos 2x} \, dx$ is equal to:

(A) $\frac{11\sqrt{3}}{2}$

(B) $8\sqrt{2} - \sqrt{\frac{3}{2}}$

(C) $8\sqrt{2} - \frac{\sqrt{3}}{2}$

(D) $\frac{22}{3}$

Ans. B

$$\begin{aligned}
 \text{Sol. } & \int_0^{\frac{11\pi}{3}} \sqrt{1 + \cos 2x} \cdot dx \\
 &= \int_0^{\frac{11\pi}{3}} \sqrt{2} |\cos x| \cdot dx \\
 &= \int_0^{3\pi} \sqrt{2} |\cos x| \cdot dx + \int_{3\pi}^{3\pi+2\pi/3} \sqrt{2} |\cos x| \cdot dx \\
 &= 3\sqrt{2} \int_0^{\pi} |\cos x| \cdot dx + \sqrt{2} \int_0^{2\pi/3} |\cos x| \cdot dx \\
 &= 6\sqrt{2} \int_0^{\pi/2} (\cos x) \cdot dx + \sqrt{2} \int_0^{\pi/2} \cos x \cdot dx - \sqrt{2} \int_{\pi/2}^{2\pi/3} \cos x \cdot dx \\
 &= 6\sqrt{2} + \sqrt{2} \frac{-\sqrt{3}}{\sqrt{2}} + \sqrt{2} \\
 &= \sqrt{2} - \sqrt{\frac{3}{2}}
 \end{aligned}$$

66. Consider the vectors $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + \hat{k}$. A parallelogram is constructed such that its diagonal are along $2\vec{a} - \vec{b}$ and $\vec{a} - 2\vec{b}$. The area of the parallelogram is:

- (A) $9\sqrt{5} \text{ sq. units}$
- (B) $\frac{9\sqrt{10}}{2} \text{ sq. units}$
- (C) $9\sqrt{10} \text{ sq. units}$
- (D) $18\sqrt{5} \text{ sq. units}$

Ans. B

$$\begin{aligned}
 \text{Sol. } & 2\vec{a} - \vec{b} = \hat{i} + 3\hat{j} - 5\hat{k} \\
 & \vec{a} - 2\vec{b} = -4\hat{i} + 4\hat{j} - 4\hat{k} \\
 & \text{Area of} = \frac{1}{2} \left| (2\vec{a} - \vec{b}) \times (\vec{a} - 2\vec{b}) \right| \\
 & (2\vec{a} - \vec{b}) \times (\vec{a} - 2\vec{b}) = 3\hat{i} + 24\hat{j} + 15\hat{k} \\
 & \text{Area} = \frac{9\sqrt{10}}{2}
 \end{aligned}$$

67. Let $\frac{\tan\left(\frac{\pi}{4}+\alpha\right)}{5} = \frac{\tan\left(\frac{\pi}{4}+\beta\right)}{3} = \frac{\tan\left(\frac{\pi}{4}+\gamma\right)}{2}$. Then $12\sin^2(\alpha-\beta) + 15\sin^2(\beta-\gamma) - 7\sin^2(\gamma-\alpha)$ is equal to?

(A) $-\frac{1}{2}$

(B) $\frac{1}{2}$

(C) 1

(D) 0

Ans. D

Sol. Taking first two

$$\frac{\tan(\pi/4 + \alpha)}{\tan(\pi/4 + \beta)} = \frac{5}{3}$$

$$\therefore \frac{\cos(\alpha + \beta)}{\sin(\alpha - \beta)} = \frac{8}{2} = 4$$

$$\therefore \cos(\alpha + \beta) = 4\sin(\alpha - \beta) \text{-----(1)}$$

Similarly

$$\cos(\beta + \alpha) = \sin(\beta - \alpha) \text{-----(2)}$$

$$3\cos(\gamma + \alpha) = -7\sin(\gamma - \alpha) \text{-----(3)}$$

After marking $\sin^{-2}$ terms

$$(1) + (2) - (3)$$

Then,

$$12\sin^2(\alpha - \beta) + 15\sin^2(\beta - \gamma) - 7\sin^2(\gamma - \alpha) = 0$$

68. The angle of intersection of the curves $y = (1+x)^{\cos x} + \sin x$ and $y = \frac{1}{2}(x^2+x+2)$ at (h,k) where h and k are integers, is θ , then $\tan \theta$ equals:

(A) $\frac{3}{4}$

(B) $\frac{4}{3}$

(C) 1

(D) 0

Ans. A

Sol. At $x = 0, y = 1$ for both curves intersect

$$\text{Now } y = (1+x)^{\cos x} + \sin x$$

$$\therefore \frac{dy}{dx} = 2 \text{ at } x = 0$$

$$\text{And } y = \frac{1}{2}(x^2 + x + 2)$$

$$\frac{dy}{dx} = \frac{1}{2} \text{ at } x = 0$$

$$\text{Now } m_1 = 2, m_2 = \frac{1}{2}$$

$$\therefore \tan \theta = \left| \frac{2 - 1/2}{1 + (2)(1/2)} \right| = \left| \frac{3/2}{2} \right| \Rightarrow \frac{3}{4}$$

69. A plane $2x + 3y + 6z = 12$ intersects the x-axis, y-axis and z-axis at A, B, C respectively. The orthocentre of the triangle ABC is:

(A) $\left(\frac{24}{49}, \frac{36}{49}, \frac{72}{49} \right)$

(B) $\left(\frac{36}{49}, \frac{72}{49}, \frac{27}{49} \right)$

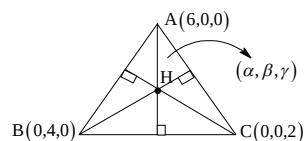
(C) $\left(\frac{24}{49}, \frac{16}{49}, \frac{1}{49} \right)$

(D) $\left(\frac{36}{49}, \frac{72}{49}, \frac{24}{49} \right)$

Ans. A

Sol. Now $AH \perp BC$

$$\text{Now } D^r \text{ of } AH = (\alpha - 6, \beta, \gamma)$$



$$D^r \text{ of } BC = (0, 4, -2)$$

Now for 1 lines,

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

$$\therefore 0(\alpha - 6) + \beta(4) - 2\gamma = 0$$

$$\therefore 4\beta - 2\gamma = 0$$

$$2\beta = \gamma \text{ --- (1)}$$

Similarly

$$-6\alpha + 2\gamma = 0 \text{ --- (2)}$$

$$-6\alpha + 4\beta = 0 \text{ --- (3)}$$

$$\therefore 2\beta = \gamma; \frac{2\beta}{3} = \alpha$$

Now may lie on

$$2x + 3y + 6z = 12$$

$$\therefore 2\alpha + 3\beta + 6\gamma = 12$$

$$2\left(\frac{2\beta}{3}\right) + 3\beta + 6(2\beta) = 12$$

$$4\beta + 9\beta + 36\beta = 36$$

$$49\beta = 36$$

$$\beta = \frac{36}{49}$$

$$\text{Now, } \alpha = \frac{24}{49} \text{ and } \gamma = \frac{72}{49}$$

$$(\alpha, \beta, \gamma) = \left(\frac{24}{49}, \frac{36}{49}, \frac{72}{49}\right)$$

70. The domain of the function given by $f(x) = \log_3 \left(\log_{\frac{1}{2}} \left(\log_2 \left(\log_{\frac{1}{3}}(x) \right) \right) \right) + \frac{x+1}{\sqrt{1-3x-4x^2}}$ is:

(A) $\left(-1, \frac{1}{3}\right)$

(B) $\left(\frac{1}{4}, \frac{1}{3}\right)$

(C) $\left(\frac{1}{9}, \frac{1}{4}\right)$

(D) $\left(\frac{1}{9}, \frac{1}{3}\right)$

Ans. C

Sol. $f(x) = \log_3 \left(\log_{1/2} \left(\log_2 \left(\log_{1/3}(x) \right) \right) \right)$

$$\frac{x+1}{\sqrt{1-3x-4x^2}}$$

$$\text{Now, } \sqrt{1-3x-4x^2} > 0$$

$$4x^2 + 3x - 1 < 0$$

$$\therefore (4x-1)(x+1) < 0$$

$$\therefore x \in (-1, 1/4)$$

$$\log_{1/2}(\log_2(\log_{1/3}(x))) > 0$$

$$\log_2(\log_{1/3}(x)) < 1$$

$$\log_{1/3}(x) < 2$$

$$x > 1/9$$

$$\text{Now } F(f(x)) \in x > \frac{1}{9} \cap x \in (-1, 1/4)$$

$$\therefore x \in \left(\frac{1}{9}, \frac{1}{4}\right)$$

71. Let α and β be the roots of the equation $x^2 - 13x - 5 = 0$. If $S_n = \alpha^n \beta + \alpha \beta^n; n \in \mathbb{N}$, then the value of $\frac{S_{13} - 13S_{12}}{S_{11}}$ is equal to ____

(A) 13

(B) -13

(C) 5

(D) -5

Ans. C

Sol.
$$\frac{S_{13} - 13S_{12}}{S_{11}}$$

$$= \frac{\alpha^{13}\beta + \alpha\beta^{13} - (\alpha + \beta)(\alpha^{13}\beta + \alpha\beta^{12})}{\alpha^{11}\beta + \alpha\beta^{11}}$$

$$= -\alpha\beta$$

$$= 5$$

72. The value of $\cot \sum_{n=1}^{\infty} \cot^{-1} \left(\frac{2}{n^2} \sum_{k=1}^n k^3 \right)$

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{1}{3}$

(D) $\frac{1}{6}$

Ans. C

Sol.
$$\begin{aligned} \cot \sum_{n=1}^{\infty} \cot^{-1} \frac{1}{n^2} \times n^2 \frac{(n+1)^2}{2} \\ &= \cot \sum_{n=1}^{\infty} \tan^{-1} \frac{2}{(n+1)^2} \\ &= \cot \sum_{h=1}^{\infty} (\tan^{-1}(n+2) - \tan^{-1} n) \\ s_n &= \begin{bmatrix} \tan^{-1} 3 - \tan^{-1} 1 \\ \tan^{-1} 4 - \tan^{-1} 2 \\ \tan^{-1}(n+2) - \tan^{-1} n \end{bmatrix} \\ s_n &= \tan^{-1} \left(\frac{3n^2 + 7n}{n^2 + 9n + 10} \right) \\ s_{\infty} &= \tan^{-1} 3 \\ \cot \tan^{-1} 3 &= \cot \cot^{-1} \frac{1}{3} \Rightarrow \frac{1}{3} \end{aligned}$$

73. The graph of the function $y = f(x)$ has a unique tangent at $(\pi^a, 0)$. If

$A = \lim_{x \rightarrow \pi^a} \frac{\ln(1 + 9f(x)) - \sin f(x)}{2f(x)}$, then $\sum_{n=1}^{\infty} A^{-n}$ is equal to:

(A) $\frac{1}{3}$

(B) $\frac{1}{2}$

(C) 4

(D) $\frac{1}{4}$

Ans. A

Sol.
$$\lim_{x \rightarrow \pi} \frac{\left\{ \frac{1}{1+9f(x)} \right\} 9f'(x) - \cos(f(x)) f'(x)}{2f'(x)}$$

$$= \frac{9-1}{2} = 4$$

$$\sum_{h=1}^{\infty} 4^{-n} = \frac{1}{4} + \frac{1}{4^2} + \dots \dots \infty$$

$$= \frac{1}{3}$$

74. Let $a_r, r=1,2,3,4$ represent four positive real numbers other than unity such that $\left(\log_e a_r, \frac{1}{\log_e a_r} \right)$ is a point on a circle whose abscissa of end of diameter are α and $-\alpha$, then the minimum value of $a_1 + a_2 + a_3 + a_4$ is:

(A) 2

(B) 3

(C) 4

(D) 5

Ans. C

Sol. Centre $(0, -f)$
 Equation of circle
 $x^2 + y^2 + 2fy + c = 0$
 $(\log_e a_r)^2 + \frac{1}{(\log_e a_r)^2} + 2f \left(\frac{1}{\log_e a_r} \right) + c = 0$
 $(\log a_r)^4 + c(\log a_r)^2 + 2f(\log a_r) = 0$
 $\log a_1, \log a_2, \log a_3, \log a_4$ are roots of equation

$$\therefore \log(a_1 \cdot a_2 \cdot a_3 \cdot a_4) = 0$$

$$a_1 a_2 a_3 a_4 = 1$$

$$\frac{a_1 + a_2 + a_3 + a_4}{4} \geq (1)^{1/4}$$

$$\therefore a_1 + a_2 + a_3 + a_4 \geq 4$$

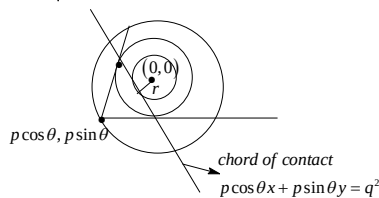
75. The chord of contact of tangents drawn from any point on the circle $x^2 + y^2 = p^2$ to the circle $x^2 + y^2 = q^2$ touches the circle $x^2 + y^2 = r^2$, then $2a - \log_e p, 4a - \log_e q, 6a - \log_e r$ are in (a is real value):

- (A) AP
(B) GP
(C) HP
(D) AGP

Ans. A

Sol. $\perp$ distance from (centre) to chord of contact

$$r = \left| \frac{p \cos \theta \times 0 + p \sin \theta \times 0 - q^2}{p} \right|$$



$$q^2 = rp$$

p, q, r are in GP

$\therefore 2a - \log_e^p, 4a - \log_e^q, 6a - \log_e^r$ are in AP

76. If t_1, t_2 and t_3 are distinct, the points $(t_1, 2at_1 + at_1^3), (t_2, 2at_2 + at_2^3)$ and $(t_3, 2at_3 + at_3^3)$ are collinear, then $\frac{4^{t_1} \cdot 6^{t_2} \cdot 8^{t_3}}{2^{t_1} \cdot 3^{t_2} \cdot 4^{t_3}}$ is equal to:

- (A) 0
(B) 4
(C) 2
(D) 1

Ans. D

Sol. *G.E. of line*

$$Ax + By + C = 0$$

$$At_1 + B(2at_1 + at_1^3) + c = 0$$

$$Bat_1^3 + (2aB + A)t_1 + c = 0$$

Sum of roots

$$t_1 + t_2 + t_3 = 0$$

$$2^{t_1} \cdot 2^{t_2} \cdot 2^{t_3} = 2^0 = 1$$

77. The area of the triangle formed by the line $x + y = 2$ and angle bisectors of the pair of straight lines $x^2 - y^2 + 2y = 1$ is a and coordinate of orthocentre of triangle is (b, c) , then $4a + b + c$ is equal to:

(A) 0

(B) 4

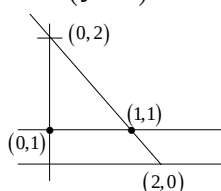
(C) 2

(D) 3

Ans. D

Sol. $x^2 - y^2 + 2y - 1 = 0$

$$x^2 - (y-1)^2 = 0$$



$$(x + y - 1)(x - y + 1) = 0$$

Angle bisector

$$\frac{x + y - 1}{\sqrt{2}} = \pm \frac{x - y + 1}{\sqrt{2}}$$

$$x + y - 1 = x - y + 1$$

$$x + y - 1 = -x + y - 1$$

$$\Rightarrow y = 1 \quad \Rightarrow x = 0$$

$$\text{Area } a = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

$$\text{Orthocentre } (0, 1) \Rightarrow b = 0, c = 1$$

$$\therefore 4a + b + c = 2 + 0 + 1 = 3$$

78. If z is the complex 5th root of unity, then $\frac{z^4}{z^4+1} + \frac{z}{1+z} + \frac{z^2}{z^2+1} + \frac{z^4}{z+z^4}$ is:

- (A) 2
(B) 0
(C) 4
(D) 1

Ans. A

Sol. $z^5 = 1$

$$\begin{aligned} & \frac{z^5}{z^5+z} + \frac{z}{z+1} + \frac{z^2}{z^2+1} + \frac{z^4}{z^2+z^5} \\ &= \frac{1}{1+z} + \frac{z}{1+z} + \frac{z^2}{z^2+1} + \frac{1}{z^2+1} \\ &= 1+1=2 \end{aligned}$$

79. If a tangent of slope 3 of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is normal to the circle $x^2 + y^2 + 4x + 1 = 0$ then the maximum value of ab is:

- (A) 2
(B) 3
(C) 4
(D) 6

Ans. D

Sol. Equation of tangent $y = 3x \pm \sqrt{9a^2 + b^2}$
This is normal to circle
 $x^2 + y^2 + 4x + 1 = 0$
Centre $(-2, 0)$
 $\Rightarrow 0 = -6 \pm \sqrt{9a^2 + b^2}$
 $36 = 9a^2 + b^2$
 $AM \geq GM$
 $\frac{9a^2 + b^2}{2} \geq \sqrt{9a^2 b^2}$
 $18 \geq 3ab$
 $6 \geq ab$

80. If $F(x) = \int_0^x f(t) \cdot dt$ and $F(x^2) = x^2(1+x)$, then value of $f(4) = n$. Then unit place of $(985697)^{n^2}$ is:

- (A) 3
(B) 7
(C) 1
(D) 9

Ans. C

Sol. $F(x) = \int_0^x f(t) dt$
 $F(x^2) = \int_0^{x^2} f(t) dt$
 $= x^2(1+x)$
 Diff. both side
 $F'(x^2) \cdot 2x = 2x + 3x^2$
 $f(x^2) = 1 + \frac{3}{2}x$
 $x=2, f(4) = 1 + 3x/2$
 $x=2, f(4) = 1 + \frac{3}{2} \times 2 = 4$
 Unit place $(985697)^{16} = 1$

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. A nine digit number is formed from 1, 2, 3, 4, 5 such that product of all digits is always 1920. The total number of ways is $393({}^9P_r)$, where the value of r is _____

Ans. 00002.00

Sol. $1920 = 5 \times 3 \times 2^7 = 5 \times 3 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = \frac{9!}{7!} = 9 \times 8$
 $1920 = 5 \times 3 \times 4 \times 2 \times 2 \times 2 \times 2 \times 2 \times 1 = \frac{9!}{5!} = 9 \times 8 \times 7 \times 6$

$$1920 = 3, 5, 4, 4, 2, 2, 2, 1, 1$$

$$\frac{9!}{2!2!3!}$$

$$3, 5, 4, 4, 4, 2, 1, 1, 1$$

$$\frac{9!}{3!3!}$$

$$\frac{9!}{7!} + \frac{9!}{5!} + \frac{9!}{2!2!3!} + \frac{9!}{3!3!}$$

$$\text{Total} = \frac{9!}{7!} + \frac{9!}{5!} + \frac{9!}{2!2!3!} + \frac{9!}{3!3!}$$

$$= \frac{9!}{7!} (1 + 42 + 210 + 140)$$

$$= 393 \cdot {}^9P_2$$

82. The sum $\sum_{i=0}^{5m} \binom{20}{i} \binom{30}{5m-i} \cdot {}^p C_q = \binom{p}{q} = 0$ if $p < q$, is maximum, then m equals _____

Ans. 00005.00

Sol. $20c_0 \times 30c_{5m} + 20c_1 \times 30c_{5m-1}$

$$+ \dots + 20c_{5m} \times 30c_0$$

$$\text{Coeff of } x^{5m} \rightarrow (1+x)^{20} (1+x)^{30}$$

$$50c_{5m}, 5m = \frac{50}{2} = 25$$

$$m = 5$$

83. Let $f(x)$ be a function such that $f(x) + f(x+2) = \sqrt{3}f(x+1)$ for all $x \in R$, then f is periodic with period T . Then value of T is _____

Ans. 00012.00

Sol. $f(x) + f(x+2) = \sqrt{3}f(x+1)$

$$x \rightarrow x+2 \text{ ----- (1)}$$

$$f(x+2) + f(x+4) = \sqrt{3}f(x+3) \text{ ----- (2)}$$

$$(1) + (2)$$

$$f(x) + 2 + (x+2) + f(x+4)$$

$$= \sqrt{3} \{ f(x+1) + f(x+3) \}$$

$$= 3f(x+2)$$

$$f(x) + f(x+4) = f(x+2)$$

$$x \rightarrow x+2 \text{ ----- (3)}$$

$$f(x+2) + f(x+6) = f(x+4) \text{ ---- (4)}$$

$$(3) + (4)$$

$$f(x) + f(x+6) = 0$$

$$f(x+6) = -f(x)$$

$$f(x+12) = -f(x+6)$$

$$= f(x)$$

$$\text{Period } T = 12$$

84. If the sum of first n terms of the A. P. 85, 90, 95, is equal to the sum of first $3n$ terms of the A. P. 9, 11, 13, ..., then $n+10$ equals _____

Ans. 00019.00

85. Let f be a continuous and differentiable function in (a, b) , if $f(x) \cdot f'(x) \geq x\sqrt{1-(f(x))^4}$ and $\lim_{x \rightarrow a^+} (f(x))^2 = 1$ and $\lim_{x \rightarrow b^-} (f(x))^2 = \frac{1}{2}$. Then the minimum value of $[a^2 - b^2] + 80$ is $([\cdot] \rightarrow \text{represent greatest integer function})$:

Ans. 00081.00

Sol. Given $\frac{f(x)f'(x)}{\sqrt{1-(f(x))^4}} \geq x$

$$\frac{2f(x)f'(x)}{\sqrt{1-(f(x))^4}} - 2x \geq 0$$

Let $g(x) = \sin^{-1}(f(x))^2 - x^2$ is non decreasing function

$$\lim_{x \rightarrow a^-} g(x) \leq \lim_{x \rightarrow b^+} g(x)$$

$$\frac{\pi}{2} - a^2 \leq \frac{\pi}{6} - b^2$$

$$a^2 - b^2 \geq \frac{\pi}{3}; [a^2 - b^2] = 1$$

86. Let $(1+x+x^2)^{20} = \sum_{k=0}^{40} a_k x^k$. If $\alpha a_{11} = \beta a_{10} + \gamma a_9$, where $\alpha, \beta, \gamma \in \mathbb{N}$ then the value of $\frac{\gamma - \alpha}{\beta}$ is _____

Ans. 00002.00

Sol. $20(1+x+x^2)^{20} (1+2x) = \sum_{k=1}^{40} (ka_k x^{k-1})(1+x+x^2)$

Equating coeff. Of x^{10} on both side

$$20[a_{10} + 2a_9] = 11a_{11} + 10a_{10} + 9a_9$$

$$\Rightarrow 11a_{11} = 10a_{10} + 31a_9$$

$$\Rightarrow \frac{\gamma - \alpha}{\beta} = \frac{31 - 11}{10} = 2$$

87. If $f(x) = \frac{3^{x-3}}{3^{1-x} + 3^x}$ for all $x \in Q$, then the value of $f\left(\frac{1}{109}\right) + f\left(\frac{2}{109}\right) + f\left(\frac{3}{109}\right) + \dots + f\left(\frac{108}{109}\right)$ is.

Ans. 00002.00

Sol. $f(x) + f(1-x) = \frac{1}{27}$

$$\frac{54}{27} = 2$$

88. Let number of non-negative integral solutions of the inequality $x + y + z \leq 6$ be m and number of natural solutions of the inequality $x + y + z + t < 10$ be n , then find the value of $\frac{2n}{m}$ is__

Ans. 00003.00

Sol. $m = {}^9C_3 \quad n = {}^9C_4$

$$\frac{2n}{m} = \frac{2 \times {}^9C_4}{{}^9C_3} = \frac{2 \times 9! \times 6! \times 3!}{4! \times 5! \times 9!} = \frac{2 \times 6}{4} = 3$$

89. Of the three independent events E_1, E_2 and E_3 , the probability that only E_1 occurs is α , only E_2 occurs is β and only E_3 occurs is γ . Let the probability p that none of events E_1, E_2 or E_3 occurs satisfy the equations $(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$. All the given probabilities are assumed to lie in the interval $(0, 1)$.

Then $\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3} =$

Ans. 00006.00

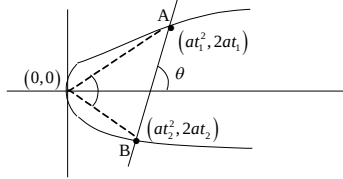
90. A line is drawn at an angle θ with positive direction of x -axis. It intersects the parabola $y^2 = 8x$ at A and B. If AB is a normal chord which subtend a right angle at the vertex of the parabola, then $\tan^2 \theta$ equals__

Ans. 00002.00

Sol. $y^2 = 8x$

Now, $t_1 t_2 = -4$

And $t_2 = -t_1 - \frac{2}{t_1}$



$$\frac{-4}{t_1} = -t_1 - \frac{2}{t_1}$$

$$\Rightarrow -t_1^2 = -2$$

$$\Rightarrow t_1^2 = 2$$

$$y^2 = 4ax$$

$$\frac{dy}{dx} = \frac{4ax}{2y}$$

$$\frac{dy}{dx} = \frac{1}{t}$$

Slope of normal $= -t$ and $\tan^2 \theta = t_1^2 = 2$